COMPUTER COMMUNICATIONS & NETWORKS — COMPREHENSIVE PRACTICE QUIZ

Targeted Review: Analog/Digital Modulation, Multiplexing (FDM, WDM, TDM), Guided Media, and Wireless Technologies

Name: _____ Student ID: _____ Date: _____

Section 1: Analog Transmission & Carrier Modulation (Lecture 5: Part 1)

Question 1.1 [1.0 Mark]

Why does a low-frequency analog information signal require modulation prior to transmission across a band-pass channel?

- ○ a. To completely eliminate all frequency variations from the original source.
- ○ b. To shift the signal spectrum into the specific passband frequency range supported by the channel.
- ○ c. To convert the continuous waveform directly into discrete binary line coding.
- ○ d. To permanently reduce the carrier center frequency down to zero Hertz.

Question 1.2 [1.0 Mark]

Which statement describes the fundamental principle of carrier modulation?

- ○ a. Maintaining all carrier characteristics completely static and constant during transmission.
- ○ b. Eliminating the high-frequency carrier wave and sending baseband pulses directly.
- ○ c. Systematically varying one or more characteristics of a high-frequency carrier wave in accordance with the information signal.
- ○ d. Multiplying the carrier frequency by zero to minimize power dissipation.

Question 1.3 [3.0 Marks]

In Amplitude Modulation (AM), a high-frequency carrier with frequency $f_c = 400\text{ kHz}$ is modulated by an audio signal having a maximum frequency $B_m = 12\text{ kHz}$. Calculate the following parameters:

1. Total bandwidth required (BW_{AM}): _____ kHz
2. Lowest frequency component of the modulated signal (f_{lower}): _____ kHz
3. Highest frequency component of the modulated signal (f_{upper}): _____ kHz

Question 1.4 [2.0 Marks]

An audio baseband signal has a bandwidth of $B_m = 8\text{ kHz}$. What is the total bandwidth required if the signal is modulated using Frequency Modulation (FM), given a modulation parameter $\beta = 4$?

Required FM Bandwidth = _____ kHz

Question 1.5 [5.0 Marks]

An Amplitude Shift Keying (ASK) signal occupies an allocated frequency band spanning from 200 kHz to 600 kHz. Assuming $d = 1$ and $r = 1$:

1. Allocated bandwidth (BW): _____ kHz
2. Midpoint frequency of the band: _____ kHz
3. Carrier frequency (f_c): _____ kHz
4. Baud rate (S): _____ kBaud
5. Resulting data rate (N): _____ kbps

Question 1.6 [3.0 Marks]

Consider an ASK signal operating across a frequency range from $4{,}000\text{ Hz}$ to $12{,}000\text{ Hz}$ configured in full-duplex transmission mode (the total spectrum is divided equally between forward and backward directions):

1. Available bandwidth for each direction: _____ kHz
2. Carrier frequency for the forward link (lower spectrum): _____ kHz
3. Carrier frequency for the backward link (upper spectrum): _____ kHz

Section 2: Digital-to-Analog Shift Keying Techniques (Lecture 5: Part 2)

Question 2.1 [2.0 Marks]

Consider a binary Frequency Shift Keying (FSK) signal transmitting at a bit rate of $N = 16\text{ kbps}$ in half-duplex mode. The two carrier frequencies are separated by 6 kHz ($f_{c1} - f_{c0} = 6\text{ kHz}$). Calculate:

1. Baud rate (S): _____ kBaud
2. Minimum required transmission bandwidth: _____ kHz

Question 2.2 [3.0 Marks]

An FSK transmission operates in full-duplex mode over a medium offering 24 kHz total bandwidth. The upper carrier frequency is configured at $f_{c1} = 36\text{ kHz}$ and the bit rate is $N = 4\text{ kbps}$. Calculate:

1. Bandwidth allocated per transmission direction: _____ kHz
2. Baud rate (S): _____ kBaud
3. Lower carrier frequency (f_{c0}): _____ kHz

Question 2.3 [8.0 Marks]

An FSK system occupies an available frequency spectrum extending from $2{,}000\text{ kHz}$ to $6{,}000\text{ kHz}$. The two carriers are symmetrically separated by $2\Delta f = (f_{c1} - f_{c0}) = 2{,}000\text{ kHz}$. Let $d = 1$:

1. Total bandwidth: _____ kHz
2. Midpoint of the bandwidth: _____ kHz
3. Lower carrier frequency (f_{c0}): _____ kHz
4. Upper carrier frequency (f_{c1}): _____ kHz
5. Carrier frequency difference $(f_{c1} - f_{c0})$: _____ kHz
6. Baud rate (S): _____ kBaud
7. Resulting data rate (N): _____ kbps

Question 2.4 [4.0 Marks]

Calculate the required transmission bandwidth (Hz) to transmit a digital bitstream of $N = 8{,}000\text{ bps}$ using the following schemes (assume $d = 1$):

1. Binary ASK: _____ Hz
2. FSK with carrier separation $2\Delta f = 6\text{ kHz}$: _____ Hz
3. QPSK ($r = 2$): _____ Hz
4. 16-QAM ($r = 4$): _____ Hz

Question 2.5 [4.0 Marks]

Determine the number of bits carried per baud ($r = \log_2 L$) for each of the following modulation constellations:

1. ASK with 16 discrete signal amplitude levels: _____ bits/baud
2. FSK with 8 distinct carrier frequencies: _____ bits/baud
3. PSK with 32 distinct constellation phase shifts: _____ bits/baud
4. QAM with a constellation consisting of 128 points: _____ bits/baud

Question 2.6 [4.0 Marks]

Calculate the resulting bit rate N (in kbps) given a transmission baud rate of $S = 4{,}000\text{ Baud/s}$ across each modulation type:

1. Binary FSK: Data rate = _____ kbps
2. Binary ASK: Data rate = _____ kbps
3. 8-PSK: Data rate = _____ kbps
4. 64-QAM: Data rate = _____ kbps

Section 3: Bandwidth Utilization & Frequency Division Multiplexing (Lecture 6: Part 1)

Question 3.1 [1.0 Mark]

Which statement best summarizes the primary objective of bandwidth utilization?

- a. Ensuring that channels exclusively transmit non-overlapping baseband square waves.
- b. The efficient and optimal allocation of available physical link bandwidth among multiple data streams.
- c. Restricting all digital communication links to asynchronous character transmission.
- d. Maximizing signal attenuation across copper conduits.

Question 3.2 [1.0 Mark]

In a multiplexed transmission system, _____ logical channel signals share the bandwidth of _____ physical transmission link.

- a. $1\text{ ; }1$
- b. $1\text{ ; }n$
- c. $n\text{ ; }n$
- d. $n\text{ ; }1$

Question 3.3 [1.0 Mark]

Which mode of signal transmission is inherently supported by individual channels within a physical link running Frequency Division Multiplexing (FDM)?

- a. Pure digital unmodulated baseband signaling
- b. Analog carrier transmission
- c. Direct memory access line pulsing
- d. Optical dispersion mapping

Question 3.4 [1.0 Mark]

Six baseband channels, each with a bandwidth of 150 kHz, are multiplexed together using FDM. What is the minimum required bandwidth of the composite link (in kHz) if a guard band of 10 kHz is required between adjacent channels to prevent crosstalk?

Total Required Link Bandwidth = _____ kHz

Question 3.5 [1.0 Mark]

A terminal transmits data at an aggregated bit rate of $N = 2.4\text{ Mbps}$ using an analog passband channel that yields a signaling symbol rate of $S = 400\text{ kBaud/s}$. Which Quadrature Amplitude Modulation scheme is deployed?

Modulation Scheme: _____ -QAM

Question 3.6 [3.0 Marks]

Four digital data channels, each transmitting at 12 Mbps, are to be multiplexed using FDM over an analog satellite transponder with an allocated bandwidth of 12 MHz. Complete the system design:

> 1. Each allocated FDM sub-channel possesses a bandwidth of: _____ MHz
>
> 2. Each 12 Mbps channel must be modulated to achieve a spectral efficiency of: _____ bits/Hz
>
> 3. The modulation technique that fulfills this requirement is: _____ (Choose: 4-QAM, 8-QAM, 16-QAM, or 64-QAM)

Question 3.7 [1.0 Mark]

An FDM system aggregates four digital channels, each operating at 6 Mbps. If 16-QAM modulation is utilized for transmission, what is the minimum theoretical bandwidth required for the composite link (in MHz)?

Required Bandwidth = _____ MHz

Question 3.8 [1.0 Mark]

True or False: Coarse Wavelength Division Multiplexing (CWDM) can support longer transmission distances than Dense Wavelength Division Multiplexing (DWDM).

- ○ True
- ○ False

Section 4: Time Division Multiplexing & Bit Padding Dynamics (Lecture 6: Part 2)

Question 4.1 [4.0 Marks]

Match the correct multiplexing paradigm (TDM, FDM, or WDM) to each operational scenario:

1. Specifically designed to combine light beams along optical fiber conduits: _____
2. Inherently partitioned into two architectural models: synchronous and statistical/asynchronous: _____
3. Used historically by analog dial-up modems: _____
4. Used by traditional terrestrial and cable television broadcasting: _____

Question 4.2 [2.0 Marks]

Eight independent 4 kbps connections are multiplexed together via TDM. Defining a base data unit as 1 bit:

1. The input bit duration is: _____ ms
2. The transmission rate of the aggregated composite link is: _____ kbps

Question 4.3 [6.0 Marks]

A synchronous TDM multiplexer combines five 40 kbps channels using a time slot allocation of 2 bits per channel. Calculate the following link attributes:

1. Frame rate (in kilo-frames per second): _____ k-frames/sec
2. Frame duration (in milliseconds): _____ ms
3. Frame duration (in microseconds): _____ μs
4. Bit rate of the aggregated composite link (in kbps): _____ kbps
5. Bit interval of the aggregated link (in milliseconds): _____ ms
6. Bit interval of the aggregated link (in microseconds): _____ μs

Question 4.4 [1.0 Mark]

In _____ , time slots are dynamically allocated on-demand to transmitting lines to prevent idle slot waste, significantly optimizing digital transmission efficiency.

- a. Synchronous TDM
- b. Statistical (Asynchronous) TDM
- c. Isochronous FDM
- d. Dense WDM

Question 4.5 [1.0 Mark]

Which of the following multiplexing/switching methodologies is NOT associated with traditional circuit switching?

- a. Frequency Division Multiplexing (FDM)
- b. Wavelength Division Multiplexing (WDM)
- c. Synchronous Time Division Multiplexing (Sync TDM)
- d. Asynchronous (Statistical) Time Division Multiplexing

Question 4.6 [6.0 Marks]

Three data sources with bit rates of 150 kbps, 450 kbps, and 180 kbps are combined using Synchronous TDM over a shared physical link. Calculate:

1. Which source channel requires bit padding? _____ kbps channel
2. What padding rate must be injected into that channel? _____ kbps
3. Total number of data bits carried in each composite frame: _____ bits
4. Resulting transmission frame rate: _____ k-frames/sec
5. Resulting frame duration: _____ μs
6. Total aggregated bit rate of the output link: _____ kbps

Section 5: Guided Media – Twisted-Pair Cabling (Lecture 7: Part 1)

Question 5.1 [1.0 Mark]

Among the twisted-pair categories listed below, which supports the highest data transmission bandwidth and throughput over standard 100 m enterprise runs?

- a. Cat3
- b. Cat4
- c. Cat5e
- d. Cat6

Question 5.2 [1.0 Mark]

True or False: A standard Ethernet Unshielded Twisted-Pair (UTP) cable consists of exactly 4 individual copper wires inside its outer jacket.

- True
- False

Question 5.3 [1.0 Mark]

According to ISO/IEC 11801 cable shielding terminology, which designation describes an Ethernet cable featuring an overall braided shield as well as individual foil shielding around each internal pair?

- a. U/UTP
- b. F/UTP
- c. SF/UTP
- d. S/FTP

Question 5.4 [1.0 Mark]

A network engineer needs to directly interconnect two router Ethernet interfaces without an intermediate switch or Auto-MDIX capability. Which cable configuration must be used?

- a. Straight-through patch cable
- b. Crossover cable
- c. Single-mode patch cord
- d. Rollover console cable

Question 5.5 [1.0 Mark]

In copper conductor cabling standards (such as American Wire Gauge - AWG), what is the relationship between the wire gauge number and signal attenuation?

- a. A smaller gauge number corresponds to a larger conductor diameter and lower signal attenuation.
- b. A smaller gauge number corresponds to a thinner conductor and higher signal attenuation.
- c. A larger gauge number guarantees higher bandwidth across long distances.
- d. Wire gauge has no influence on line resistance or signal attenuation.

Question 5.6 [1.0 Mark]

Which modular connector standard is universally utilized for 8-conductor UTP Ethernet networking cables?

- a. RJ-11
- b. RJ-45
- c. RG-58
- d. BNC

Question 5.7 [1.0 Mark]

True or False: Increasing the twist density (more twists per foot) results in worse signal quality and increased crosstalk in UTP cables.

- True
- False

Question 5.8 [1.0 Mark]

Which operational trade-off is most characteristic of Shielded Twisted-Pair (STP) compared to Unshielded Twisted-Pair (UTP) cabling in noisy industrial plant environments?

- a. STP requires proprietary hardware, but eliminates all installation labor.
- b. STP provides superior immunity against electromagnetic noise and crosstalk, but is bulkier, stiffer, and more expensive to install.
- c. STP eliminates the need for wire twisting altogether.
- d. STP is immune to all physical wear, but cannot exceed 10 meters in distance.

Question 5.9 [1.0 Mark]

Which commercial twisted-pair wiring pinout standard is most predominantly deployed across structured cabling installations in Thailand?

- a. TIA/EIA 568A
- b. TIA/EIA 568B
- c. ISO/IEC 27001
- d. IEEE 802.3ah

Question 5.10 [1.0 Mark]

What physical principle explains why insulated copper conductors are twisted around each other in pairs inside an Ethernet cable?

- a. To improve the cable's electrical breakdown voltage limit.
- b. To ensure equal physical length and prevent tangling during pulling.
- c. To cancel out differential electromagnetic interference (EMI) and reduce crosstalk.
- d. To convert high-frequency electrical signals directly into optical guided waves.

Section 6: Guided Media – Coaxial & Fiber Optic Cabling (Lecture 7: Part 2)

Question 6.1 [1.0 Mark]

What standardized prefix designation identifies coaxial cable ratings (e.g., RG-6, RG-58, RG-59)?

Answer: _____

Question 6.2 [1.0 Mark]

Between optical fiber and standard copper coaxial cable, which medium exhibits significantly higher signal attenuation per kilometer?

- a. Optical Fiber
- b. Coaxial Cable

Question 6.3 [1.0 Mark]

Which optical fiber classification possesses the smallest core diameter (typically around 8 to 10 micrometers)?

- a. Multimode, step index
- b. Multimode, graded index
- c. Single mode
- d. Plastic optical fiber

Question 6.4 [1.0 Mark]

Which fiber optic refractive profile features the highest optical density at the center of the core that gradually decreases toward the core-cladding boundary?

- a. Multimode, graded index
- b. Multimode, step index
- c. Single mode, step index
- d. Homogeneous cladding guide

Question 6.5 [1.0 Mark]

Which of the following connectors is NOT a standard termination hardware for coaxial cables?

- a. BNC
- b. SMA
- c. RJ-45
- d. Type N

Question 6.6 [1.0 Mark]

Under which condition does a propagating light wave undergo total internal reflection (TIR) within the core of an optical fiber?

- a. When the incident angle is smaller than the critical angle.
- b. When the incident angle is strictly greater than the critical angle.
- c. When the refractive index of the cladding exceeds that of the core.
- d. When the incident angle is exactly 0° relative to the normal.

Section 7: Unguided Media & Wireless Technologies (Lecture 8)

Question 7.1 [2.0 Marks]

Rank the following electromagnetic wave regions in order of increasing frequency (lowest frequency first): Radio waves, Infrared, Light waves, Microwaves.

Lowest: _____ < _____ < _____ < Highest: _____

Question 7.2 [1.0 Mark]

Which IEEE 802.11 Wi-Fi standard operates in the millimeter-wave 60 GHz frequency band and possesses the smallest communication range (in-room LOS)?

- a. IEEE 802.11b/g/n
- b. IEEE 802.11ah (HaLow)
- c. IEEE 802.11ad (WiGig)
- d. IEEE 802.11af (White-Fi)

Question 7.3 [1.0 Mark]

Which IEEE 802.11 Wi-Fi standard operates in the sub-GHz television white space spectrum and provides the largest physical coverage area?

- a. IEEE 802.11ad
- b. IEEE 802.11af
- c. IEEE 802.11a/ac
- d. IEEE 802.11g

Question 7.4 [2.0 Marks]

Which long-range, sub-GHz wireless RF modulation technology is classified as a prominent Low Power Wide Area Network (LPWAN) architecture for IoT?

Answer: _____

Question 7.5 [2.0 Marks]

Which broadband wireless standard (IEEE 802.16) is formally classified as a Wireless Metropolitan Area Network (WMAN) technology?

Answer: _____

Question 7.6 [1.0 Mark]

Select all unlicensed (license-free) frequency spectrum bands utilized by commercial IEEE 802.11 Wi-Fi networks:

- [ ] a. 2.4 GHz (ISM)
- [ ] b. 3.0 GHz (VHF)
- [ ] c. 5.0 GHz (UNII)
- [ ] d. 5.0 GHz (ISM)
- [ ] e. 6.0 GHz (UHF)

Question 7.7 [1.0 Mark]

In which operational mode does a Wi-Fi client station connect to enterprise networks and the Internet via a centralized Access Point (AP)?

- ○ a. Independent BSS (Ad-hoc) Mode
- ○ b. Peer-to-Peer (P2P) Direct Mode
- ○ c. Infrastructure Mode
- ○ d. Promiscuous Monitoring Mode

Question 7.8 [1.0 Mark]

What defines a Stealth Access Point in wireless LAN security terminology?

- ○ a. An access point running high-gain directional antennas to evade local signal sweepers.
- ○ b. An access point that suppresses broadcasting its SSID in management beacon frames.
- ○ c. A rogue AP that injects duplicate frames into neighbor BSS networks.
- ○ d. An access point operating exclusively on classified military frequencies.

Question 7.9 [4.0 Marks]

An IEEE 802.11g access point operates within the _____ band, employing non-overlapping channels such as _____ . Conversely, an IEEE 802.11ac access point operates in the _____ band, supporting channels such as _____ .

Question 7.10 [1.0 Mark]

What core operational capability does MIMO (Multiple-Input Multiple-Output) introduce into modern Wi-Fi systems (such as 802.11ax / Wi-Fi 6)?

- ○ a. Utilizing multiple antennas to transmit and receive multiple parallel spatial data streams simultaneously.
- ○ b. Switching channel frequency rapidly to evade interference from cordless appliances.
- ○ c. Encrypting MAC payload headers across multiple cryptographic ciphers.
- ○ d. Doubling device transmission voltage during low-activity intervals.

Question 7.11 [3.0 Marks]

Classify the standard antenna radiation pattern (Omnidirectional or Unidirectional/Directional) typically paired with each wireless application:

> 1. Standard indoor office Wi-Fi Access Point: _____ antenna
> 2. Point-to-point terrestrial microwave relay link: _____ antenna
> 3. Infrared (IrDA) transceiver port: _____ antenna

Question 7.12 [9.0 Marks]

Select the corresponding electromagnetic propagation mode (Ground-wave propagation, Sky-wave propagation, or Line-of-Sight (LOS) propagation) for each wireless scenario:

> 1. Commercial AM Radio broadcast ($535\text{ kHz} - 1605\text{ kHz}$): _____ mode
> 2. High-Frequency (HF) amateur / Ham radio ionospheric refraction: _____ mode
> 3. Commercial FM Radio broadcast ($88\text{ MHz} - 108\text{ MHz}$): _____ mode
> 4. 4G LTE and 5G cellular communication: _____ mode
> 5. Television Infrared (IR) remote control signaling: _____ mode
> 6. Geostationary satellite microwave transponder links: _____ mode

Answer Key & Step-by-Step Solutions

Section 1: Analog Transmission & Modulation Solutions

1.1 Correct Answer: b. To shift the signal spectrum into the frequency range supported by the channel. Baseband information cannot propagate through band-pass filters without up-conversion.

1.2 Correct Answer: c. Systematically varying one or more characteristics of a high-frequency carrier wave according to the information.

1.3 Solutions:

```
1. BW_AM = 2 * B_m = 2 * 12 kHz = 24 kHz
2. f_lower = f_c - B_m = 400 kHz - 12 kHz = 388 kHz
3. f_upper = f_c + B_m = 400 kHz + 12 kHz = 412 kHz
```

1.4 Solution:

```
Carson's Rule: BW_FM = 2(1 + beta) * B_m
With beta = 4: BW_FM = 2(1 + 4) * 8 kHz = 10 * 8 kHz = 80 kHz
```

1.5 Solutions:

```
1. Bandwidth = 600 kHz - 200 kHz = 400 kHz
2. Midpoint = (200 + 600) / 2 = 400 kHz
3. Carrier frequency (f_c) = Midpoint = 400 kHz
4. BW = (1 + d) * S => 400 = (1 + 1) * S => S = 200 kBaud
5. Data rate (N) = S * r = 200 * 1 = 200 kbps
```

1.6 Solutions:

```
1. Bandwidth per direction = (12,000 Hz - 4,000 Hz) / 2 = 4,000 Hz = 4 kHz
2. Forward band = 4 kHz to 8 kHz => f_c1 = (4 + 8) / 2 = 6 kHz (or 6,000 Hz)
3. Backward band = 8 kHz to 12 kHz => f_c2 = (8 + 12) / 2 = 10 kHz (or 10,000 Hz)
```

Section 2: Digital-to-Analog Shift Keying Solutions

2.1 Solutions:

```
1. In binary FSK, r = 1 => Baud rate S = N = 16 kBaud
2. BW = (f_c1 - f_c0) + S = 6 kHz + 16 kBaud = 22 kHz
```

2.2 Solutions:

```
1. Bandwidth per direction = 24 kHz / 2 = 12 kHz
2. Baud rate S = N = 4 kBaud
3. BW_dir = S + (f_c1 - f_c0) => f_c0 = S + f_c1 - BW_dir = 4 + 36 - 12 = 28 kHz
```

2.3 Solutions:

```
1. Total Bandwidth = 6,000 kHz - 2,000 kHz = 4,000 kHz
2. Midpoint = (2,000 + 6,000) / 2 = 4,000 kHz
3. Carrier separation 2*delta_f = 2,000 kHz => delta_f = 1,000 kHz
f_c0 = 4,000 - 1,000 = 3,000 kHz
4. f_c1 = 4,000 + 1,000 = 5,000 kHz
5. Separation = 5,000 - 3,000 = 2,000 kHz
6. BW = (1 + d)*S + 2*delta_f => 4,000 = (1 + 1)*S + 2,000 => 2S = 2,000 => S =
1,000 kBaud
7. Data rate N = S = 1,000 kbps (1 Mbps)
```

2.4 Solutions (N = 8,000 bps, d = 1):

```
1. ASK: S = N = 8,000 => BW = (1 + 1)*8,000 = 16,000 Hz
2. FSK: BW = (1 + 1)*8,000 + 6,000 = 16,000 + 6,000 = 22,000 Hz
3. QPSK: r = 2 => S = 8,000 / 2 = 4,000 => BW = 2 * 4,000 = 8,000 Hz
4. 16-QAM: r = 4 => S = 8,000 / 4 = 2,000 => BW = 2 * 2,000 = 4,000 Hz
```

2.5 Solutions:

```
Formula: r = log2(L)
1. 16-ASK: log2(16) = 4 bits/baud
2. 8-FSK: log2(8) = 3 bits/baud
3. 32-PSK: log2(32) = 5 bits/baud
4. 128-QAM: log2(128) = 7 bits/baud
```

2.6 Solutions (S = 4,000 Baud/s):

```
Formula: N = r * S
1. Binary FSK: r = 1 => N = 1 * 4,000 = 4 kbps
2. Binary ASK: r = 1 => N = 1 * 4,000 = 4 kbps
3. 8-PSK: r = log2(8) = 3 => N = 3 * 4,000 = 12 kbps
4. 64-QAM: r = log2(64) = 6 => N = 6 * 4,000 = 24 kbps
```

Section 3: Bandwidth Utilization & FDM Solutions

3.1 Correct Answer: b. The efficient and optimal allocation of available physical link bandwidth among multiple data streams.

3.2 Correct Answer: d. $n\text{ ; }1$ (n logical channels share 1 physical link).

3.3 Correct Answer: b. Analog transmission (FDM partitions continuous spectral bandwidth).

3.4 Solution:

```
BW_total = (N * BW_ch) + ((N - 1) * BW_guard)
BW_total = (6 * 150 kHz) + ((6 - 1) * 10 kHz) = 900 + 50 = 950 kHz
```

3.5 Solution:

```
r = N / S = 2,400 kbps / 400 kBaud = 6 bits/baud
Constellation points L = 2^r = 2^6 = 64 => 64-QAM
```

3.6 Solutions:

```
1. Sub-channel BW = 12 MHz / 4 = 3 MHz
2. Bit density = 12 Mbps / 3 MHz = 4 bits/Hz (or bits/baud)
3. Required Modulation: 2^4 = 16-QAM
```

3.7 Solution:

```
Total data rate = 4 * 6 Mbps = 24 Mbps
16-QAM carries log2(16) = 4 bits/baud
Minimum Bandwidth = 24 Mbps / 4 bits/baud = 6 MHz (or 6 Mbauds/s)
```

3.8 Correct Answer: False. DWDM supports substantially longer distances and far denser optical wavelength packing than CWDM.

Section 4: Time Division Multiplexing Solutions

4.1 Answers: 1. WDM | 2. TDM | 3. FDM | 4. FDM

4.2 Solutions:

```
1. Bit duration = 1 / 4 kbps = 0.25 ms (250 microseconds)
2. Aggregated link rate = 8 * 4 kbps = 32 kbps
```

4.3 Solutions:

```
1. Frame rate = 40 kbps / 2 bits = 20 k-frames/sec
2. Frame duration (ms) = 1 / 20k = 0.05 ms
3. Frame duration (usec) = 0.05 * 1,000 = 50 usec
4. Aggregated bit rate = 5 * 40 kbps = 200 kbps
5. Bit interval (ms) = 1 / 200 kbps = 0.005 ms
6. Bit interval (usec) = 0.005 * 1,000 = 5 usec
```

4.4 Correct Answer: b. Statistical (Asynchronous) TDM.

4.5 Correct Answer: d. Asynchronous (Statistical) TDM is packet/buffer oriented, not a circuit-switched static reservation method.

4.6 Solutions:

```
Base channel rate = 150 kbps
Ch1: 150 / 150 = 1 slot
Ch2: 450 / 150 = 3 slots
Ch3: 180 / 150 = 1.2 => Non-integer! Must pad to next multiple: 300 kbps (2 slots)
1. Channel requiring padding: 180 kbps channel
2. Padding rate needed = 300 - 180 = 120 kbps
3. Bits per frame = 1 + 3 + 2 = 6 bits (assuming 1 bit/slot)
4. Frame rate = base rate = 150 k-frames/sec
5. Frame duration = 1 / 150k = 6.67 usec (or 10/1.5 usec)
6. Aggregated link bit rate = 150 k-frames/sec * 6 bits/frame = 900 kbps
```

Section 5: Guided Media – Twisted-Pair Cabling Solutions

5.1 Correct Answer: d. Cat6 (Supports up to 250 MHz and 10 Gbps over shorter runs, 1 Gbps up to 100 m).

5.2 Correct Answer: False. A standard UTP cable contains 4 twisted pairs, totaling 8 copper wires.

5.3 Correct Answer: d. S/FTP (Overall braid screen 'S', with foil 'FTP' around each individual pair).

5.4 Correct Answer: b. Crossover cable (used for like-device connections: router-to-router, switch-to-switch, PC-to-PC).

5.5 Correct Answer: a. Smaller gauge number represents a thicker wire with less electrical resistance and lower attenuation.

5.6 Correct Answer: b. RJ-45 (8P8C modular connector).

5.7 Correct Answer: False. Higher twist density increases electromagnetic noise cancellation, improving cable quality and reducing crosstalk.

5.8 Correct Answer: b. STP offers superior noise/crosstalk immunity, but is significantly bulkier, less flexible, and requires grounding.

5.9 Correct Answer: b. TIA/EIA 568B is the dominant standard in Thailand and commercial networks.

5.10 Correct Answer: c. Twisting ensures equal exposure to external noise, enabling differential cancellation of EMI and crosstalk.

Section 6: Guided Media – Coaxial & Fiber Optic Solutions

6.1 Correct Answer: RG (Radio Guide).

6.2 Correct Answer: b. Coaxial Cable (attenuation in copper coax is orders of magnitude higher per kilometer than in low-loss optical fiber).

6.3 Correct Answer: c. Single mode (~8 to 10 µm core diameter).

6.4 Correct Answer: a. Multimode, graded index (core index of refraction follows a parabolic curve from center to edge).

6.5 Correct Answer: c. RJ-45 (an 8-pin connector for twisted pair, not coaxial). Coax uses BNC, N, SMA, and F-type.

6.6 Correct Answer: b. Total Internal Reflection occurs when the light travels from higher to lower refractive index and the incident angle exceeds the critical angle.

Section 7: Unguided Media & Wireless Solutions

7.1 Correct Sequence: Radio waves < Microwaves < Infrared < Light waves.

7.2 Correct Answer: c. IEEE 802.11ad (Operates at 60 GHz; millimetric waves experience high atmospheric absorption, yielding in-room range).

7.3 Correct Answer: b. IEEE 802.11af (White-Fi operates in sub-GHz TV White Spaces, penetrating obstacles over long distances).

7.4 Correct Answer: LoRa (or LoRaWAN).

7.5 Correct Answer: WiMAX (IEEE 802.16).

7.6 Correct Answers: a, c, d. 2.4 GHz (ISM), 5.0 GHz (UNII), and 5.0 GHz (ISM) are unlicensed bands. (VHF/UHF bands listed are broadcast/licensed).

7.7 Correct Answer: c. Infrastructure Mode.

7.8 Correct Answer: b. An access point that hides its presence by suppressing its SSID from beacon frames.

7.9 Correct Answers: Band: **2.4 GHz**; Channels: **1, 6, 11**; Band: **5 GHz**; Channels: **36, 40, 44, 48 (or 42, 58, 106)**.

7.10 Correct Answer: a. MIMO uses multiple antennas to transmit and receive multiple spatial streams simultaneously.

7.11 Correct Answers: 1. Omnidirectional | 2. Unidirectional (Directional) | 3. Unidirectional (Directional).

7.12 Propagation Modes:

- 1. Commercial AM Radio: **Ground-wave propagation** (follows Earth curvature)
- 2. Ham Radio (HF): **Sky-wave propagation** (ionospheric reflection)
- 3. FM Radio: **Line-of-Sight (LOS) propagation**
- 4. 4G/5G Cellular: **Line-of-Sight (LOS) propagation**
- 5. TV Remote (Infrared): **Line-of-Sight (LOS) propagation**
- 6. Satellite Microwave: **Line-of-Sight (LOS) propagation**